Credit Portfolio Risk Analysis

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Portfolios: Portfolio 1 and Portfolio 2

Course: Introduction to Credit Risk and Applications in Python

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1. Introduction

This report analyzes the credit risk of two loan portfolios. The objective is to clean up the chaotic raw data, establish a credit risk model, simulate the one-year portfolio loss, and compare the risks of Portfolio 1 and Portfolio 2.

The analysis follows four main steps. First, I clean the raw data and merge it into modeling tables. Second, I prepare the model inputs, including EAD, LGD, PD, and the correlation parameter. Third, I use a one-factor model to simulate the one-year loss distribution and calculate risk measures, including expected loss, VaR, and ES. Fourth, I interpret the results and compare the risk of Portfolio 1 and Portfolio 2.

2. Data Cleaning and Preparation

The raw data consists of entities_info.csv for firm characteristics, portfolio_1.csv and portfolio_2.csv for portfolio holdings, stock_returns.csv for firm-level returns, industry_index_returns.csv for industry-level returns, and default_history_20y.csv for historical default records. I cleaned each dataset before using it for modeling.

First, I standardized all entity codes. Some entity codes had inconsistent formats, such as missing hyphens or lower-case letters. I used a helper function to remove spaces, convert codes to upper case, and add the missing hyphen after the first letter when needed. This was important because the entity code is the key variable used to merge the datasets. Otherwise, the same firm may not be matched correctly across files.

Second, I standardized the industry labels. Different versions of the same industry name were converted into a consistent format, such as “Industry A” and “Industry B”.

Third, I handled missing values. Missing LGD and base PD values were first filled with the median value within the same industry. If missing values still remained, I used the overall median. Missing EAD values in the portfolio files were filled using the median EAD of the same portfolio. Missing stock return values were filled using time-series interpolation within each firm. Missing industry index return values were filled using time-series interpolation within each industry.

Fourth, I converted all date columns into a consistent datetime format.

Fifth, I removed duplicate rows, including repeated observations for the same entity and date.

After cleaning, I merged the portfolio holdings with the cleaned firm information table. This created two modeling tables, one for each portfolio. I also combined them into one table with a portfolio label.

The final modeling table contains the main variables needed for the model: entity code, EAD, booking date, industry, region, LGD, base_pd_hint, and portfolio name.

These cleaned variables are then used as inputs for the credit risk model in the next section.

3. Model Setup and Methodology

3.1 Model Inputs

The model uses EAD, LGD, PD, and the asset correlation parameter as its main inputs.

For PD, I use the cleaned base_pd_hint as the one-year probability of default. I use this variable because it is already provided at the obligor level and directly matches the PD input needed in the simulation model. This keeps the model simple and avoids adding another estimation step.

For LGD, I use the cleaned LGD value from the firm information table. For EAD, I use the cleaned exposure value from the portfolio files. These two variables determine the loss amount if an obligor defaults.

The individual expected loss is calculated as:

$$EL_i = EAD_i \cdot LGD_i \cdot PD_i$$

3.2 Correlation Parameter

For the asset correlation parameter, I use the Basel II corporate correlation formula from the Fitch report discussed in class (Fitch Ratings, 2008). The formula is:

$$\rho_i = 0.24 - 0.12 \cdot \left(\frac{1 - e^{-50PD_i}}{1 - e^{-50}}\right)$$

The formula gives an asset correlation value based on each obligor’s PD. I use it because the portfolios contain corporate credit exposures, and the formula is made for this type of credit risk model. The asset correlation is not directly given in the raw data, so I use this formula as a clear approximation.

The Basel II formula gives asset correlation, not the factor loading directly. In the one-factor model, the loading on the common factor is the square root of the asset correlation.

Using the Basel II formula is a modeling assumption. The method is simple, easy to reproduce, and uses each obligor’s PD. However, the functional form and coefficients come from the Basel II formula, not from a direct estimation of the dependence structure in the sample data. Therefore, it may not fully capture portfolio-specific dependence.

3.3 Simulation Method

Using the cleaned modeling table, I set up a one-factor latent variable model to simulate correlated defaults. In this type of model, each obligor has a latent credit quality variable. Default occurs when this variable falls below the default threshold.

I use repeated random simulations to generate the one-year portfolio loss distribution. I run 10,000 simulations for each portfolio. For each simulation, I generate one systematic risk factor and obligor-specific shocks. The systematic factor represents a common economic condition that can affect many obligors at the same time. The idiosyncratic shock represents firm-specific risk. The latent variable is:

$$X_i = \sqrt{\rho_i}\, F + \sqrt{1 - \rho_i}\, \varepsilon_i$$

Because the Basel II formula gives asset correlation, I use the square root of rho_i as the factor loading.

Default occurs when the latent variable is below the default threshold:

$$X_{\mathrm{i}} \leq \Phi^{-1}(PD_{\mathrm{i}})$$

If an obligor defaults, the loss is:

$$Loss_i = EAD_i \cdot LGD_i$$

The total portfolio loss in each simulation is the sum of all default losses. From the simulated loss distribution, I calculate simulated expected loss, VaR, and ES.

4. Results and Discussion

4.1 Parameter Summary

The parameter summary is taken from part2_parameter_summary.csv. Table 1 reports the main input parameters and expected loss for the two portfolios.

Table 1. Parameter summary for Portfolio 1 and Portfolio 2

Portfolio	Obligors	Total EAD	Avg. PD	Avg. LGD	Avg. rho	EL	EL ratio
Portfolio 1	30	676.4m	2.00%	47.1%	16.6%	6.13m	0.91%
Portfolio 2	30	761.6m	3.43%	50.3%	14.3%	13.24m	1.74%

Both portfolios have 30 obligors, but Portfolio 2 has larger total EAD and higher credit risk inputs. Its average PD and LGD are higher than those of Portfolio 1. As a result, Portfolio 2 has a much higher expected loss and expected loss ratio. This already suggests that Portfolio 2 is riskier.

4.2 Simulation Results and Risk Measures

The risk measures are taken from part3_risk_measures.csv. Table 2 reports the simulated expected loss, VaR, ES, and mean loss ratio for the two portfolios.

Table 2. Simulated risk measures for Portfolio 1 and Portfolio 2

Portfolio	Simulated EL	VaR 95%	VaR 99%	ES 95%	ES 99%	Mean loss ratio
Portfolio 1	6.22m	29.27m	49.88m	42.21m	63.98m	0.92%
Portfolio 2	13.46m	50.53m	79.62m	69.37m	101.89m	1.77%

The simulated expected losses are close to the expected losses calculated from EAD, LGD, and PD. This result is reasonable. The average simulated loss is expected to be close to the model expected loss.

The results show that Portfolio 2 has higher risk than Portfolio 1. It has higher simulated expected loss, higher VaR, and higher ES. This means that Portfolio 2 has both higher average loss and higher tail loss.

4.3 Loss Distribution

Figure 1 shows the simulated loss distributions of the two portfolios.

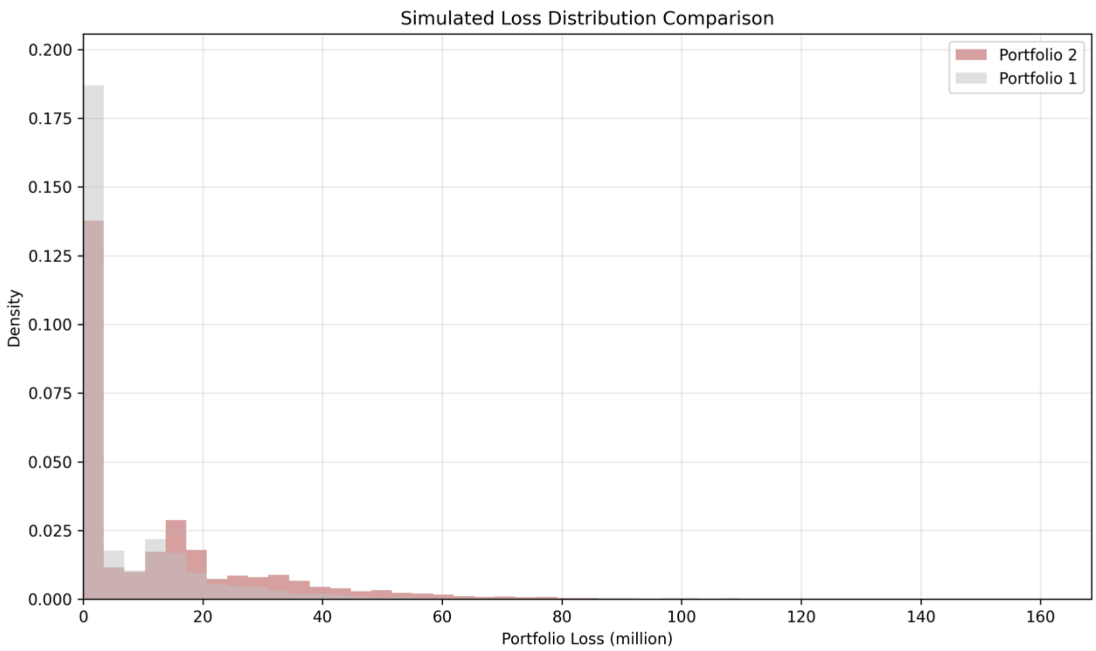

Figure 1. Simulated loss distribution comparison for Portfolio 1 and Portfolio 2

The simulated loss distributions are right-skewed. Most simulated losses are close to zero or relatively small. However, both portfolios have a long right tail.

This pattern is common in credit risk. In most simulations, only a few obligors default. In bad scenarios, several obligors may default at the same time, so portfolio losses can become much larger. Portfolio 2 has a longer and heavier right tail than Portfolio 1. This means that Portfolio 2 has a higher chance of large losses. This is consistent with the VaR and ES results in the previous section.

4.4 Interpretation and Limitations

Overall, the results show that Portfolio 2 is riskier than Portfolio 1. The main reasons are its higher total EAD, higher average PD, and higher average LGD. These factors increase both expected loss and tail risk.

The one-factor model is useful because it allows defaults to be dependent. This is more realistic than assuming that all defaults are independent. In bad scenarios, the common systematic factor can make several obligors default at the same time.

However, the model also has some limitations. First, PD is taken from the cleaned base_pd_hint, so I do not estimate a new PD model. Second, LGD is treated as fixed. In reality, LGD may change, especially in stress periods. Third, the Basel II correlation formula is a modeling assumption. It is simple and clear, but it may not fully capture portfolio-specific dependence.

Even with these limitations, the model gives a clear comparison between the two portfolios and is suitable for this assignment.

5. Conclusion

This report compares the credit risk of Portfolio 1 and Portfolio 2. After cleaning the data, I set up a one-factor model using EAD, LGD, PD, and the Basel II corporate asset correlation formula.

The results show that Portfolio 2 is riskier than Portfolio 1. Portfolio 2 has higher simulated expected loss, higher VaR, and higher expected shortfall. The loss distribution also shows a heavier right tail for Portfolio 2.

Overall, Portfolio 2 should be viewed as the riskier portfolio.

References

Fitch Ratings. (2008). *Basel II correlation values: An empirical analysis of EL, UL and the IRB model*. Fitch Ratings.